严格性总表（测试）

本页仅用于测试 `master_table_zh.tex` 的编译（见表 1–4）。

表 1　QCD 因子化相关结论的严格性总表（一）：基础与“严格核心”（条目 1–6）。标签含义见正文；“关键文献”栏只列最关键的已核实来源。

#	主题	已确立的结论（要点）	关键文献	标签	主要缺口 / 反例
1	Wilson OPE、光锥 OPE	重整化微扰论中的短距 OPE（ϕ^4：渐近且逐圈收敛）；纯 Yang–Mills 中 OPE 在规范不变算符上闭合；电流乘积的光锥展开	[1, 2, 3, 4, 5, 6]	A 全阶论证（ϕ^4、纯 YM：R 严格；非微扰：C 猜想）	含夸克 QCD：仅断言“与其他理论同样证明”（Collins 1984 §12.8）；IJJSZ 2018：“widely assumed”
2	DIS 矩与 DGLAP	$M_N = \sum_i C_{N,i} A_{N,i} + O(\Lambda^2/Q^2)$；算符定义 $\overline{\text{MS}}$ PDF 的 DGLAP；$\gamma_{ij}(N) = \int_0^1 dz\, z^{N-1} P_{ij}$；核函数已知至三圈	[7, 8, 9, 10, 11]	R 严格（矩：以条目 1 为前提）	逐矩的 $O(1/Q^2)$ 余项不给出对 x 一致的控制（$x \to 1$）
3	红外有限性	$R_{e^+e^-}$ 逐阶有限；无质量理论欧氏非例外动量格林函数红外有限；IR 安全喷注截面有限	[12, 13, 14, 15, 16, 17]	R 严格（总截面、欧氏）；A 全阶论证（喷注）	有色初态 Bloch–Nordsieck：V 破坏（幂次压低）[18, 19]；KLN 需初末态求和及前向散射 [20]
4	Landau 方程、Coleman–Norton	夹点奇异的必要（物理区亦充分）条件；经典传播解释	[21, 22, 23, 24]	A 全阶论证	教科书证明不完整（Collins 2011 §5.4.3）；无质量情形的完整证明仅见预印本
5	幂次计数、主导区域	幂律 $Q^p(\lambda/Q)^{\alpha(R)} \times (\lambda_S/Q)^{\beta(R)}$；主导区域目录；Glauber 夹点判据	[15, 25, 26, 27, 28]	A 全阶论证	Feynman 规范下主导区域不完整（附加不连通硬散射，对图求和后抵消 [29]）；重叠区域需减除
6	DIS 共线因子化（图方法）	x 空间 $F = \sum_i C_i \otimes f_i + O((\Lambda/Q)^a)$，PDF 为规范不变算符矩阵元	[30, 31, 32, 33, 27, 34]	A 全阶论证	非局域算符的 BRST 分类未证（Collins 2011 §11.3.2 明确假定）；K 胶子余项 [35]

表 2　QCD 因子化相关结论的严格性总表（二）：PDF 定义、强子–强子过程与 TMD（条目 7–10）。

#	主题	已确立的结论（要点）	关键文献	标签	主要缺口 / 反例
7	算符定义 PDF/FF；支撑；正定性	规范不变定义与重整化；$\lvert\xi\rvert \le 1$，$f_q(-\xi) = -f_{\bar q}(\xi)$；裸密度的数密度解释	[10, 36, 27]	R 严格（定义、重整化、支撑）；C 猜想（$\overline{\text{MS}}$ 正定性）	正定性之争：[37] 对 [38]，及 [39]（约 $Q^2 \gtrsim 5\,\text{GeV}^2$ 以上，半定量）
8	Drell–Yan 与 Glauber 抵消	包含性 DY 首阶幂次因子化；旁观者 Glauber 交换经对切割求和与围道变形后抵消	[40, 41, 42, 27, 43]	A 全阶论证	Collins 2011 §14.4.8 “遗留问题”；SCET 完整证明未见发表 [44]；喷注否决/E_T 类观测量 V 破坏 [45, 46]；m^4/s^2 阶失效 [47]
9	强子对撞大 p_T 单举强子/喷注、重夸克	唯象中普遍假定的共线因子化（含碎裂函数或喷注函数）	[48, 49, 50]	C 猜想	有色末态的 Glauber 抵消未证 [43]；Collins 2011 §14.7；N3LO 质量奇异问题 [51]
10	TMD 因子化（DY、SIDIS、e^+e^-）	b 空间因子化；Collins–Soper 与 RG 演化；小 b OPE 至共线 PDF	[52, 53, 54, 55, 56, 57, 58]	A 全阶论证	DY 依赖条目 8 的 Glauber 论证；SCET 推导在论证后舍弃 Glauber 拉氏量 [59]

表 3　QCD 因子化相关结论的严格性总表（三）：过程依赖、破坏与推广（条目 11–20）。

#	主题	已确立的结论（要点）	关键文献	标签	主要缺口 / 反例
11	Sivers 函数变号	$f_{1T}^{\perp}\big\|_{\mathrm{DY}} = -f_{1T}^{\perp}\big\|_{\mathrm{SIDIS}}$（Boer–Mulders 同）	[60, 61, 62, 63]	R 严格（以 TMD 因子化为前提）	实验一致但非决定性：[64, 65]；全局拟合尚不能排除同号 [66]
12	TMD 因子化破坏	背对背强子产生中标准与“广义”TMD 因子化均失效	[67, 68, 69]	V 破坏	模型中的显式反例（足以否定一般定理）
13	类空共线因子化破坏、超领头对数、Glauber 算符	类空共线极限的严格因子化在圈图层次失效；超领头对数；SCET Glauber 算符	[51, 70, 71, 72, 44]	V 破坏（严格共线因子化）；C 猜想（NNLO 以上质量奇异因子化）	三圈 active–active Glauber 使间隙截面在否决标度以下与 PDF 因子化相容 [73]
14	衍射因子化	衍射 DIS 因子化成立；强子–强子硬衍射不成立	[74, 75, 76, 77]	A 全阶论证（DDIS）；V 破坏（$p\bar{p}$、分解光生）	CDF：衍射结构函数相对 HERA 预言压低 $O(10)$；H1 光生约 0.5
15	双部分子散射	双 DY：全阶 Glauber 抵消（共线与 TMD）；SPS/DPS 重复计数解决；软胶子脱耦；求和规则	[78, 43, 79, 80, 81]	A 全阶论证（双 DY）；C 猜想（有色末态）	软矩阵 S 演化与正定性无一般证明 [82]；MPI 敏感观测量 [45]
16	硬遍举过程	π 形状因子首阶幂次；DVCS；纵向光子介子电产生；横向光子端点失效	[83, 84, 85, 86, 87]	A 全阶论证；V 破坏（γ_T^*）	可及 Q^2 下端点与软重叠主导 [88]；Sudakov/k_T 方案为模型 [89]
17	SCET 因子化证明	算符层次软–共线脱耦；$B \to D\pi$ 全阶；DIS/DY/DVCS 重推；双喷注/thrust	[90, 91, 92, 93, 94, 95]	A 全阶论证	假定模式完备与 Glauber 模缺席或抵消；Collins 2011 §15.9 的保留意见
18	无粲 B 衰变 QCD 因子化（BBNS）	首阶幂次在 $O(\alpha_s)$ 与两圈（NNLO）验证；$B \to D\pi$ 两圈证明	[96, 97, 98, 99, 100]	P 部分	非全阶证明（BBNS 2000 自述）；湮灭与扭度 3 旁观者幂次修正端点发散
19	NRQCD 因子化	衰变 $\Gamma = \sum_n 2\,\mathrm{Im} f_n \times \langle O_n\rangle / M^{d_n-4}$；产生 $\sigma = \sum_n F_n \times \langle O_n^H\rangle / M^{d_n-4}$	[101, 102, 48, 103]	A 全阶论证（衰变）；C 猜想（产生；NNLO 检验 P 部分）	八重态产生算符需 Wilson 线；更高阶或需新非局域算符；普适性张力 [104]
20	小 x：k_T 因子化、BFKL、BK/JIMWLK	LL 高能因子化；LL/NLL BFKL；LL BK/JIMWLK（及 NLO BK）	[105, 106, 107, 108, 109]	P 部分（k_T）；A 全阶论证（LL/NLL 内）；C 猜想（饱和）	非积分胶子分布过程依赖 [110]；无受控幂次修正

表 4　QCD 因子化相关结论的严格性总表（四）：高扭度、方法论与其他（条目 21–30）。

#	主题	已确立的结论（要点）	关键文献	标签	主要缺口 / 反例
21	高扭度与扭度 3	DIS 扭度 4（OPE）；强子 $1/Q^2$ 广义因子化；扭度 3 单自旋不对称	[111, 36, 112, 113, 114]	R 严格（DIS 矩）；A 全阶论证（$1/Q^2$）；P 部分（扭度 3）；V 破坏（强子 $1/Q^4$）	与首阶扭度的重整子混合；扭度 3 无全阶证明
22	重整子	IR 重整子对应幂次修正；微扰级数阶乘发散	[115, 116, 117, 118]	A 全阶论证；C 猜想（非微扰 OPE）	QCD 中无非微扰的 OPE 定义
23	区域方法	欧氏极限展开严格；闵氏阈值、在壳、Glauber 展开仅在特例中证明	[119, 120, 121, 122, 28]	R 严格（欧氏）；P 部分（闵氏）	一般情形为猜想 [122]
24	LaMET 与 pseudo-PDF	准 PDF 与 pseudo-ITD 对光锥 PDF 的匹配；Wilson 线夸克双线性算符全阶乘法可重整	[123, 124, 125, 126, 127]	A 全阶论证	依赖 QCD 类空 OPE；幂次修正 $\sim \Lambda^2/(x^2(1-x)P_z^2)$ [128]；批评 [129]
25	阈值重求和与再因子化	首阶幂次阈值因子化与 Mellin 空间指数化；次首阶幂次 LL	[130, 131, 132, 133]	A 全阶论证（LP）；P 部分（NLP）	NLP 端点发散需逐例处理 [134]
26	非全局对数	$O(\alpha_s^2)$ 起的 NGL；BMS 方程（大 N_c）；有限 N_c；EFT 因子化	[135, 136, 137, 138]	R 严格（存在性）；A 全阶论证（大 N_c LL）；P 部分	强子对撞中 Glauber 相位导致超领头对数（条目 13）
27	DIS 重夸克方案	保留重夸克质量的因子化，余项一致为 $O(\Lambda/Q)$；ACOT 等方案	[139, 140, 34, 141, 142]	A 全阶论证	强子–强子中保留入射夸克质量不可取（Collins 2011 §14.6）
28	光生与分解光子	光子结构函数（点状部分）；直接光生类 DIS；分解光生类强子–强子；质子光子 PDF	[143, 144, 145, 146]	A 全阶论证（直接）；C 猜想（分解）；V 破坏（衍射分解）	直接/分解在 LO 以上的划分不唯一；衍射光生压低 [77, 147]
29	核 PDF 与核中因子化	eA 同 DIS；pA DY 同 DY 加 $A^{1/3}/Q^2$ 核增强扭度 4	[112, 148, 149, 150]	A 全阶论证	要求能量远大于随靶长度增长的标度 [149]；小 x 饱和区 C 猜想
30	其他要点	$e^+e^- \to hX$ 碎裂；Sudakov 形状因子；多腿振幅红外结构；振幅层次硬–软–共线因子化	[10, 151, 152, 153, 154, 155]	A 全阶论证	偶极公式作为全阶陈述在三圈失效（V 破坏）；Feige–Schwartz 排除初末态共线构型

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